

# NUNNA SAI TEJA

Electronics and Communication Engineering  
KL University, Guntur

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## Summary

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Electronics and Communication Engineering graduate with hands-on exposure to Embedded Systems, C/C++ programming, microcontrollers, sensor interfacing, motor control, and hardware prototyping. Experienced with Arduino, ESP32, ESP8266, and STM32 platforms through academic projects and an Embedded System Developer Virtual Internship. Familiar with GPIO, ADC, PWM, UART, SPI, and I2C concepts, along with hardware/software debugging, circuit testing, and system validation. Strong interest in Embedded Software and Embedded Systems development.

## Technical Skills

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- **Programming:** Embedded C, C++, Python, Java
- **Microcontrollers:** Arduino, ESP32, ESP8266, STM32, TM4C123GH6PM
- **Embedded Interfaces:** GPIO, ADC, PWM, UART, SPI, I2C, Timers
- **Hardware:** Sensors, Relays, Motors, Motor Drivers, Switches, LEDs, Power Modules, Basic Electronic Circuits
- **Embedded Systems:** Sensor Interfacing, Motor Control, Hardware Debugging, Software Debugging, Circuit Testing, Hardware Prototyping
- **Tools:** STM32CubeIDE, Arduino IDE, Altium Designer, EasyEDA, MATLAB, Linux/Ubuntu
- **Other:** IoT, MQTT, PCB Design, Serial Communication, System Testing and Documentation

## Experience

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### Embedded System Developer Virtual Intern

Jan 2025 – Mar 2025

*AICTE – EduSkills Virtual Internship, Supported by Microchip*

- Completed a 10-week Embedded System Developer Virtual Internship focused on embedded system development and microcontroller-based applications.
- Gained practical exposure to embedded programming concepts, microcontrollers, hardware interfacing, and embedded system development.
- Worked with embedded-system concepts involving sensors, GPIO-based control, communication interfaces, and system-level integration.
- Applied embedded development concepts through technical assignments, practical activities, testing, and system-level problem solving.

## Projects

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### Surveillance Car – ESP32-CAM Embedded System

July 2024 – Dec 2024

*Embedded C/C++, ESP32-CAM, Motor Control, GPIO*

- Developed a Wi-Fi-controlled embedded vehicle using ESP32-CAM with motor driver, DC motors, LEDs, camera module, and control inputs.
- Implemented GPIO-based control for motors and peripheral devices and tested hardware responses under different operating conditions.
- Integrated motors, camera, LEDs, and other electronic modules through circuit wiring, power connections, and signal interfacing.
- Debugged motor, wiring, power, and software issues through systematic hardware and functional testing.

### Smart Agricultural Monitoring System

Jan 2024 – Apr 2024

*ESP32, Embedded IoT, Sensor Interfacing, ADC*

- Developed an embedded monitoring system using ESP32 with soil-moisture, temperature, and humidity sensors.
- Interfaced sensor signals with the microcontroller and implemented sensor data acquisition and processing.
- Performed sensor calibration, hardware testing, fault isolation, and system validation under varying environmental conditions.
- Troubleshoot sensor connections and software logic to improve system reliability and accuracy of real-time readings.

## Smart Home Automation System

ESP8266, C/C++, Relays, GPIO, MQTT

Aug 2023 – Dec 2023

- Developed a Wi-Fi-based home automation system using ESP8266, relay modules, switches, and connected loads.
- Implemented GPIO-based relay control and configured MQTT communication for remote device operation.
- Assembled and tested the prototype circuit, verified signal connections, and performed functional testing of individual modules.
- Troubleshoot wiring, relay, communication, and software issues during system integration.

## Education

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### B.Tech in Electronics and Communication Engineering

Koneru Lakshmaiah University, Guntur

Aug 2022 – 2026

CGPA: 9.53

### Intermediate (MPC)

Aditya Junior College, Kakinada

2020 – 2022

Marks: 933

### Secondary Education

Gamyam English Medium School

2010 – 2020

Marks: 580

## Certifications

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- **Microchip Embedded System Developer** (EDUSKILLS – AICTE) – View
- **Embedded System and IoT Programming** (TESSOLVE – Level 1) – View
- **IoT Programming and Automation** (TESSOLVE – Level 2) – View
- **Altium PCB Designing Certification** – View
- **NPTEL Programming in Java Certification** – View

## Positions of Responsibility

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- **Core Member, Always@VLSI** – Conducted hands-on sessions related to EDA tools and electronics design.
- **Member, PULSE ECE Body** – Participated in IoT, embedded systems workshops, technical activities, and team projects.

## Interests

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- Embedded Systems and Firmware Development
- Microcontroller Programming and Hardware Interfacing
- Electronics Prototyping and Circuit Debugging